

Functorial Kummer Lifts Preserve Gauge Triviality: A Three-Channel Obstruction for the Homogeneous Hénon Route

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Abstract

The homogeneous cubic Hénon phase gives a natural scalar scaling cocycle, but that cocycle is an exact gauge coboundary. We test the smallest proposed nonscalar escape: the direct three-channel Kummer grading. We prove a general representation-stability theorem: every finite-dimensional functor applied to a scalar coboundary is again a coboundary, and therefore has identity holonomy on every closed orbit and every repetition. The result covers tensor products, exterior powers, duals, parity shifts, and virtual supertraces. A type-strict exact certificate checks 10,985 component identities in a thirteen-state model and an independent eleven-state replay. Thus a direct $\mathbb{Z}/3$ -graded lift cannot supply new prime weights or a Hénon determinant. A surviving successor must add nonfunctorial channel permutation or projective central-extension data.

1 Introduction

The area-preserving Hénon program begins with a nonlinear symplectic map and asks whether its periodic, arithmetic, or quantized structures can enter a Hilbert–Pólya-style determinant. The foundational map and its exact generating-function structure are described in [2]. In the adelic scaling realization, the homogeneous cubic phase $\phi(x) = \psi(2x^3)$ produces the cocycle

$$c(a, x) = \phi(ax)\phi(x)^{-1}.$$

The scalar line is gauge trivial. A tempting response is to retain the three cubic characters as a graded vector bundle. This paper shows that this change is insufficient whenever it is performed functorially.

Our contribution is one sharp statement. *Representations do not turn a coboundary into monodromy.* This elementary fact has a consequential application: every direct Kummer channel, every finite tensor construction, and every associated virtual supertrace has identity closed holonomy. The obstruction is all-orbit and independent of a prime cutoff.

The paper distinguishes three evidence levels. The representation theorem is **PROVED**. The finite programs are **NUMERICALLY_CERTIFIED** implementation checks. The possibility of a nonfunctorial monomial or projective extension remains **OPEN**.

2 The homogeneous cubic cocycle

Let a group G act on a set X on the left, and let A be a group. For a map $\phi : X \rightarrow A$, define

$$c(g, x) = \phi(gx)\phi(x)^{-1}. \tag{1}$$

Then

$$c(gh, x) = c(g, hx)c(h, x).$$

In the Hénon application, X is the nonzero adelic scaling site, G is the scaling group, and A is the circle group generated by the global additive character. The phase $2x^3$ is fixed by the source map; no prime or Riemann-zero data enter its definition. Tate harmonic analysis supplies the ambient normalization [1].

Let $\mu_3 = \langle \zeta \rangle$. The direct three-channel model records the characters $1, \chi, \chi^2$. If the scalar cocycle is represented by a μ_3 -valued label, its fibre map is

$$U(g, x) = \text{diag}(1, c(g, x), c(g, x)^2). \quad (2)$$

The determinant of (2) is one, but determinant one alone would not exclude nontrivial matrix holonomy. The theorem below does.

3 Representation-stable gauge triviality

Theorem 3.1 (Functorial coboundary theorem). *Let $G \curvearrowright X$, let A be a group, and let c be given by (1). For every representation $\rho : A \rightarrow \text{GL}(V)$, the represented cocycle satisfies*

$$\rho(c(g, x)) = F(gx)F(x)^{-1}, \quad F = \rho \circ \phi.$$

For a closed word $g_n \cdots g_1$ based at x , its chronological holonomy on V is the identity.

Proof. The first identity follows by applying the homomorphism ρ to (1). Write $x_0 = x$ and $x_j = g_j x_{j-1}$. Chronological multiplication gives

$$\prod_{j=n}^1 F(x_j)F(x_{j-1})^{-1} = F(x_n)F(x_0)^{-1}.$$

If the word closes, $x_n = x_0$, so the product is the identity. □

Corollary 3.2 (Finite tensor closure). *The conclusion of Theorem 3.1 survives finite direct sums, tensor products, duals, exterior and symmetric powers, parity shifts, and formal differences in the representation ring.*

Proof. Each listed construction is a functor on finite-dimensional representations. It sends $F(gx)F(x)^{-1}$ to the corresponding quotient of the transformed gauges. Formal differences alter traces and determinants but not the identity of each closed holonomy. □

For the three Kummer characters, take $F(x) = \text{diag}(1, \phi(x), \phi(x)^2)$. Equation (2) is precisely $F(gx)F(x)^{-1}$. Thus its full matrix holonomy, not only its determinant, is one.

4 Prime repetitions and determinant consequences

Let C_p denote an inherited scaling-site primitive orbit with length $\log p$. The r -fold repetition has holonomy I by Theorem 3.1. Hence the Hénon increment to any orbit trace is just the dimension or superdimension of a fixed representation. It does not depend on p , on chronology, or on the cubic dynamics.

Proposition 4.1 (No new functorial prime weight). *Any orbit Euler factor obtained solely by applying a finite representation functor to the scalar homogeneous cocycle is identical to the corresponding factor for the undecorated scaling system, up to a fixed multiplicity and an exactly removable gauge.*

Proof. All closed holonomies equal the identity. Every character evaluation is therefore its value at the identity, which is independent of the orbit. A graded character replaces the dimension by a superdimension but introduces no Hénon-dependent coefficient. $\square$

This proposition does not say that all three-channel cocycles are trivial. It says that those obtained functorially from the scalar phase are trivial. A monomial cocycle of the form

$$F(gx)P(g)F(x)^{-1}$$

can retain the channel permutation $P(g)$. Such data are not generated by changing representation category alone and form the next candidate.

5 Exact finite certificate

The certificate works in $X = \mathbb{Z}/13\mathbb{Z}$ with translation action and phase label

$$f(x) = x^3 + 2x + 1 \pmod{3}.$$

It exhausts all (g, h, x) , verifies the cocycle identity, checks the three represented gauges, and checks the graded determinant. A closed path with increments $(2, 5, 1, 5)$ starts at 4 and has channel exponents $(0, 0, 0)$. The totals are shown in Table 1.

Table 1: Exact implementation checks.

Check family	Count
Cocycle identities	2,197
Represented gauge identities	6,591
Graded determinant identities	2,197
Independent eleven-state replay	exhaustive

The computation is not used to infer the theorem. It detects sign, order, and channel-index errors in the implementation. The independent replay uses a different modulus and phase function.

6 Route-A and Route-B audit

The primitive scaling orbits and $\log p$ clock remain exact, so A1 is weak rather than absent. A2 fails: there is no new zeta or determinant. The grading is analytically coherent but inherits all divisor data, giving only partial A3. The quantization is natural, giving A4. The resulting tuple is

$$(A1_{\text{WEAK}}, A2_{\text{FAIL}}, A3_{\text{PARTIAL ANALYTIC STRUCTURE}}, A4_{\text{NATURAL QUANTIZATION}}).$$

Route B is not authorized because Route A is not ready and no candidate operator domain or spectral determinant has been constructed.

7 Conclusion

The direct cubic Kummer lift does not cross the bridge left by C37. Functorial operations preserve the exact gauge, so every closed holonomy is the identity at the matrix level. The obstruction is reusable well beyond this model: whenever a proposed nonscalar lift is obtained only by applying a representation functor to a scalar coboundary, it cannot create periodic monodromy. The smallest honest successor must add intrinsic permutation or projective data and then face a separate all-prime determinant test.

References

- [1] John Tate. Fourier analysis in number fields and hecke's zeta-functions. *Thesis, Princeton University*, 1950.
- [2] Liang Wang. An area-preserving h enon-map model, 2026. Repository manuscript.